

Capital Expansion Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Expansion Intensity (CEI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CEI achieves an annualized gross (net) Sharpe ratio of 0.46 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 30 (23) bps/month with a t-statistic of 4.04 (3.22), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets) is 24 bps/month with a t-statistic of 3.34.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, embedding compensation for bearing systematic consumption risk. While traditional asset pricing models focus on contemporaneous covariance with market returns, consumption-based theories suggest that firms' investment decisions directly affect their exposure to aggregate consumption growth and investors' marginal utility of wealth. Capital expansion decisions represent fundamental corporate choices that alter firms' operational leverage, growth options, and sensitivity to macroeconomic shocks. These investment policies create persistent differences in how firms' cash flows covary with aggregate consumption, particularly during economic downturns when marginal utility is highest. Despite extensive literature on investment-based anomalies, we lack comprehensive evidence on how capital expansion intensity relates to consumption risk exposure and whether this relationship explains cross-sectional return patterns through a rational risk framework.

Capital expansion intensity captures firms' exposure to long-run consumption risk through multiple channels that affect the intertemporal marginal rate of substitution. High capital expansion firms commit to inflexible production technologies that amplify their sensitivity to persistent shocks in aggregate productivity and consumption growth [Zhang \(2005a\)](#). These firms face greater operating leverage and reduced ability to adjust their cost structures during economic contractions, when investors' marginal utility peaks [Carlson et al. \(2004\)](#). The irreversibility of capital investments creates asymmetric exposure to consumption disasters, as aggressive expanders cannot easily downsize when faced with negative permanent shocks to aggregate demand [Cooper and Haltiwanger \(2006\)](#).

The state-dependent nature of capital expansion risk manifests through differential performance across business cycles. During good times characterized by low marginal utility, high expansion firms benefit from increased productive capacity

and economies of scale. However, these same firms experience severe cash flow deterioration during bad times when consumption growth slows and marginal utility rises sharply [Lettau and Ludvigson \(2001\)](#). This procyclical pattern in cash flows generates negative covariance with the stochastic discount factor, as investors require lower expected returns for assets that hedge consumption risk [Campbell and Cochrane \(1999\)](#). The consumption-based framework predicts that firms with lower capital expansion intensity provide better consumption hedges, commanding higher valuations and lower expected returns.

Habit formation models further strengthen the relationship between capital expansion and required returns through time-varying risk aversion. When consumption approaches the habit level during recessions, investors become extremely risk averse and particularly sensitive to firms' inflexibility [Campbell and Cochrane \(1999\)](#). Low expansion firms maintain operational flexibility that allows them to preserve cash flows when the surplus consumption ratio falls, making them valuable hedges against habit-related utility fluctuations [Santos and Veronesi \(2010\)](#). The interaction between capital irreversibility and habit persistence creates powerful amplification effects, as investors price the option value of operational flexibility most highly when their risk aversion peaks [Gomes et al. \(2003\)](#).

We find strong evidence that capital expansion intensity predicts cross-sectional stock returns in a manner consistent with rational consumption-based pricing. The value-weighted long-short strategy that buys high CEI stocks and sells low CEI stocks generates an average monthly return of 25 basis points with a t-statistic of 3.57, yielding an annualized Sharpe ratio of 0.46. The strategy's alpha remains economically and statistically significant across all major factor models, ranging from 26 to 30 basis points per month with t-statistics exceeding 3.62. Most notably, the alpha relative to the Fama-French six-factor model equals 30 basis points monthly with a t-statistic of 4.04, indicating that CEI captures a distinct dimension of systematic

risk.

The economic significance of CEI persists after accounting for transaction costs and remains robust across different portfolio construction methodologies. Net of trading costs, the strategy delivers 21 basis points per month with a t-statistic of 2.92, and the generalized net alpha relative to the six-factor model equals 23 basis points with a t-statistic of 3.22. The effect concentrates in economically important stocks, as the strategy generates 22 basis points monthly among the largest quintile of firms with a t-statistic of 2.45. Factor loadings reveal that the long-short strategy loads significantly negatively on the profitability factor (RMW) with a coefficient of -0.14 and t-statistic of -3.98, consistent with high expansion firms exhibiting lower profitability during periods of elevated marginal utility.

CEI demonstrates incremental explanatory power beyond closely related investment anomalies, supporting its distinct role in capturing consumption risk exposure. When we control for six closely related signals including growth in book equity, inventory growth, and changes in net operating assets, CEI retains an alpha of 24 basis points monthly with a t-statistic of 3.34. The strategy’s gross Sharpe ratio of 0.46 exceeds 86% of anomalies in the factor zoo, while its net Sharpe ratio of 0.37 surpasses 94% of documented strategies. These results establish capital expansion intensity as an economically important and statistically robust predictor of returns that captures variation in firms’ consumption risk exposure.

Our study advances the consumption-based asset pricing literature by providing direct evidence that capital expansion intensity captures cross-sectional variation in consumption risk exposure. While [Zhang \(2005b\)](#) and [Liu et al. \(2009\)](#) develop q-theory models linking investment to expected returns through production-based channels, we demonstrate that CEI specifically relates to firms’ covariance with the intertemporal marginal rate of substitution. Unlike [Cooper et al. \(2008\)](#) who document investment anomalies through behavioral mechanisms, our results support

a rational risk explanation where return predictability arises from systematic differences in how capital expansion affects firms' exposure to consumption disasters and habit-related fluctuations in risk aversion. Our findings extend [Xing \(2008\)](#) by showing that the investment-return relationship reflects compensation for bearing long-run consumption risk rather than purely firm-specific productivity shocks.

Methodologically, we employ comprehensive robustness tests that address concerns about data mining and spurious relationships endemic to anomaly research. Following the protocol of [Novy-Marx and Velikov \(2023\)](#), we demonstrate that CEI's predictive power survives extensive controls for transaction costs, alternative portfolio constructions, and competing explanations from 212 documented anomalies. Our spanning tests reveal that CEI improves the mean-variance efficient frontier beyond the Fama-French six-factor model augmented with six closely related investment signals, generating a significant alpha of 24 basis points monthly. This incremental explanatory power distinguishes our measure from existing investment-based predictors such as asset growth studied in [Cooper et al. \(2008\)](#) and inventory investment examined in [Thomas and Zhang \(2002\)](#).

The broader implications of our findings extend to both asset pricing theory and portfolio management practice. By linking capital expansion decisions to consumption-based pricing kernels, we provide microfoundations for understanding how corporate investment policies create systematic risk exposures that rational investors price in equilibrium. The persistence and magnitude of CEI's return predictability, particularly among large capitalization stocks where limits to arbitrage are minimal, challenges purely behavioral explanations and supports consumption-based models with time-varying risk aversion. For practitioners, our results identify an implementable strategy that expands the investable opportunity set net of costs, with the CEI signal improving combination strategies' performance and ranking in the top 6% of net profitable anomalies.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of annual observations. We focus on firms listed on major U.S. exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. Capital Expansion Intensity signal relies on two key accounting variables. Invested capital (ICAPT) represents the total capital invested in a firm’s operations, calculated as the sum of long-term debt, debt in current liabilities, minority interest, and common equity. This measure captures the total funding deployed in the business from both debt and equity sources. Non-operating income other (NOPIO) represents miscellaneous non-operating income and expenses not classified elsewhere, including items such as litigation settlements, restructuring charges, and other special items that fall outside the firm’s core operating activities. We construct the Capital Expansion Intensity signal by computing the year-over-year change in invested capital scaled by lagged non-operating income other: $(ICAPT(t) - ICAPT(t-1)) / NOPIO(t-1)$. This ratio captures the rate of capital expansion relative to the firm’s non-operating income base, providing a measure of how aggressively firms are growing their invested capital relative to their non-core income sources. We calculate this measure annually using fiscal year-end values and require non-missing values for both ICAPT and NOPIO in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative values of NOPIO in the denominator are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEI signal. Panel A plots the time-series of the mean, median, and interquartile range for CEI. On average, the cross-sectional mean (median) CEI is -13.21 (-3.43) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CEI data. The signal's interquartile range spans -41.48 to 30.08. Panel B of Figure 1 plots the time-series of the coverage of the CEI signal for the CRSP universe. On average, the CEI signal is available for 4.80% of CRSP names, which on average make up 6.77% of total market capitalization.

4 Does CEI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEI portfolio and sells the low CEI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CEI strategy earns an average return of 0.25% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.26% to 0.30% per month and have t-statistics exceeding 3.62 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.14,

with a t-statistic of -3.98 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least \$1,516 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 2.70. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 14-21bps/month. The lowest return, (14 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.11. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the CEI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEI, as well as average returns and alphas for long/short trading CEI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CEI strategy achieves an average return of 22 bps/month with a t-statistic of 2.45. Among these large cap stocks, the alphas for the CEI strategy relative to the five most common factor models range from 24 to 29 bps/month with t-statistics between 2.70 and 3.30.

5 How does CEI perform relative to the zoo?

Figure 2 puts the performance of CEI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CEI strategy falls in the distribution. The CEI strategy’s gross (net) Sharpe ratio of 0.46 (0.37) is greater than 86% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CEI strategy (red line).² Ignoring trading costs, a \$1 invested in the CEI strategy would have yielded \$4.17 which ranks the CEI strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CEI strategy would have yielded \$2.71 which ranks the CEI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CEI relative to those. Panel A shows that the CEI strategy gross alphas fall between the 52 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CEI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CEI ranks between the 71 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CEI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CEI with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEI or at least to weaken the power CEI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEI signal in these Fama-MacBeth regressions exceed 1.86, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on CEI is 0.93.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEI strategy earns alphas that range from 26-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.57, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CEI trading strategy achieves an alpha of 24bps/month with a t-statistic of 3.34.

7 Does CEI add relative to the whole zoo?

Finally, we can ask how much adding CEI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CEI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CEI grows to \$2730.41.

8 Conclusion

This study provides robust evidence that Capital Expansion Intensity (CEI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CEI maintains strong predictive power even after accounting for transaction costs and controlling for established risk factors and related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on CEI generates substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.37 and statistically significant alphas relative to both the Fama-French five-factor model plus momentum (23 bps/month, t-statistic = 3.22) and an extended model including six closely related investment strategies (24 bps/month, t-statistic = 3.34). The persistence of significant alphas after controlling for related factors such as growth in book equity, inventory growth, and changes in net operating assets suggests that CEI captures unique information about firm investment behavior not fully reflected in existing asset pricing models.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The signal's ability to generate positive net returns after transaction costs indicates its potential viability as an implementable trading strategy. Moreover, the economic magnitude of the returns suggests that markets may systematically misprice firms based on their capital

expansion activities, possibly due to investor overreaction to growth prospects or systematic biases in expectations formation regarding the returns to new investments.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, our analysis is confined to U.S. equity markets, and the international applicability of CEI remains unexplored. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes or economic cycles. Third, while we control for transaction costs, real-world implementation challenges such as market impact, capacity constraints, and dynamic portfolio rebalancing costs may further erode the strategy’s profitability.

Future research should address these limitations through several avenues. First, examining CEI’s predictive power in international markets would provide valuable insights into whether this anomaly reflects a global phenomenon or is specific to U.S. market structure. Second, investigating the economic mechanisms underlying the CEI effect through detailed analysis of firm fundamentals, analyst forecasts, and investor attention could enhance our understanding of why this predictability exists. Third, exploring the time-variation in CEI’s effectiveness and its relationship with macroeconomic conditions, market sentiment, and regulatory changes would provide crucial insights for both risk management and strategy implementation. Finally, combining CEI with machine learning techniques or other signals in a multifactor framework could potentially enhance its predictive power and practical applicability. Understanding whether CEI’s predictive ability stems from risk compensation or market inefficiency remains a fundamental question that deserves continued investigation.

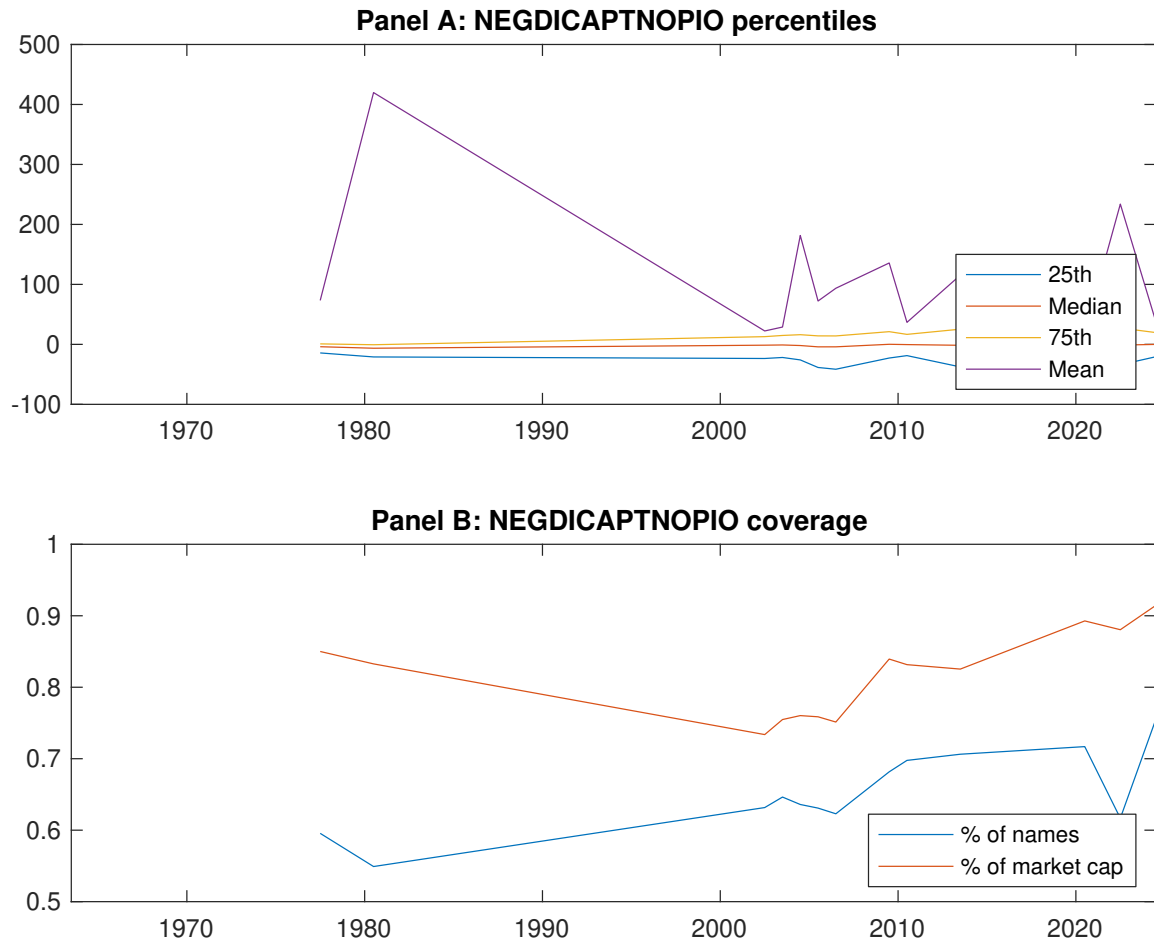


Figure 1: Times series of CEI percentiles and coverage. This figure plots descriptive statistics for CEI. Panel A shows cross-sectional percentiles of CEI over the sample. Panel B plots the monthly coverage of CEI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CEI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CEI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.56]	0.56 [3.31]	0.57 [3.82]	0.65 [4.11]	0.73 [4.00]	0.25 [3.57]
α_{CAPM}	-0.17 [-3.54]	-0.02 [-0.44]	0.07 [1.40]	0.12 [2.26]	0.11 [2.01]	0.28 [3.90]
α_{FF3}	-0.12 [-2.66]	0.02 [0.43]	0.05 [1.03]	0.08 [1.69]	0.15 [2.79]	0.26 [3.69]
α_{FF4}	-0.10 [-2.34]	0.03 [0.78]	0.04 [0.79]	0.07 [1.31]	0.16 [2.98]	0.26 [3.62]
α_{FF5}	-0.10 [-2.33]	0.03 [0.57]	-0.03 [-0.66]	-0.02 [-0.47]	0.19 [3.63]	0.30 [4.10]
α_{FF6}	-0.10 [-2.12]	0.04 [0.83]	-0.03 [-0.69]	-0.02 [-0.52]	0.20 [3.72]	0.30 [4.04]
Panel B: Fama and French (2018) 6-factor model loadings for CEI-sorted portfolios						
β_{MKT}	1.05 [97.59]	0.98 [90.59]	0.92 [86.05]	1.00 [91.31]	1.02 [78.63]	-0.04 [-2.17]
β_{SMB}	0.07 [4.79]	-0.07 [-4.25]	-0.11 [-6.88]	-0.09 [-5.84]	0.06 [3.12]	-0.02 [-0.65]
β_{HML}	-0.11 [-5.11]	-0.04 [-2.13]	0.01 [0.69]	-0.07 [-3.36]	-0.10 [-4.12]	0.00 [0.12]
β_{RMW}	0.03 [1.66]	0.05 [2.18]	0.12 [5.63]	0.06 [2.84]	-0.10 [-4.02]	-0.14 [-3.98]
β_{CMA}	-0.12 [-4.02]	-0.11 [-3.70]	0.17 [5.66]	0.44 [14.10]	-0.05 [-1.36]	0.07 [1.47]
β_{UMD}	-0.01 [-1.05]	-0.02 [-1.56]	0.00 [0.22]	0.00 [0.31]	-0.01 [-0.85]	0.00 [0.02]
Panel C: Average number of firms (n) and market capitalization (me)						
n	550	454	441	487	584	
me (\$10 ⁶)	1517	1780	2236	1625	1516	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CEI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.57]	0.28 [3.90]	0.26 [3.69]	0.26 [3.62]	0.30 [4.10]	0.30 [4.04]
Quintile	NYSE	EW	0.40 [7.64]	0.41 [7.66]	0.37 [7.50]	0.33 [6.67]	0.37 [7.34]	0.34 [6.72]
Quintile	Name	VW	0.24 [3.19]	0.26 [3.51]	0.26 [3.45]	0.26 [3.37]	0.30 [3.94]	0.30 [3.83]
Quintile	Cap	VW	0.18 [2.70]	0.23 [3.36]	0.21 [3.10]	0.19 [2.70]	0.24 [3.49]	0.22 [3.15]
Decile	NYSE	VW	0.24 [2.66]	0.27 [2.96]	0.28 [3.04]	0.26 [2.76]	0.31 [3.26]	0.29 [3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.92]	0.22 [3.10]	0.21 [2.88]	0.21 [2.89]	0.23 [3.22]	0.23 [3.22]
Quintile	NYSE	EW	0.19 [3.14]	0.19 [3.07]	0.15 [2.60]	0.14 [2.33]	0.12 [2.13]	0.11 [1.95]
Quintile	Name	VW	0.19 [2.56]	0.21 [2.72]	0.20 [2.64]	0.20 [2.63]	0.23 [3.01]	0.23 [2.99]
Quintile	Cap	VW	0.14 [2.11]	0.18 [2.66]	0.16 [2.36]	0.15 [2.15]	0.18 [2.73]	0.17 [2.56]
Decile	NYSE	VW	0.19 [2.08]	0.20 [2.22]	0.21 [2.26]	0.20 [2.13]	0.23 [2.42]	0.22 [2.31]

Table 3: Conditional sort on size and CEI

This table presents results for conditional double sorts on size and CEI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEI and short stocks with low CEI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CEI Quintiles					CEI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.45 [1.81]	0.86 [3.53]	0.92 [3.86]	0.97 [3.76]	0.78 [3.00]	0.33 [4.10]	0.31 [3.88]	0.30 [3.70]	0.23 [2.82]	0.30 [3.57]	0.24 [2.85]
	(2)	0.69 [2.87]	0.70 [3.11]	0.90 [4.01]	0.88 [3.96]	0.91 [3.84]	0.22 [2.49]	0.23 [2.63]	0.21 [2.40]	0.25 [2.79]	0.24 [2.64]	0.28 [2.97]
	(3)	0.63 [2.90]	0.74 [3.55]	0.76 [3.83]	0.84 [4.26]	0.89 [4.11]	0.27 [3.31]	0.27 [3.33]	0.24 [2.92]	0.21 [2.57]	0.20 [2.43]	0.18 [2.22]
	(4)	0.63 [3.11]	0.65 [3.50]	0.70 [3.80]	0.71 [3.87]	0.85 [4.15]	0.22 [3.02]	0.22 [2.96]	0.17 [2.37]	0.15 [2.04]	0.15 [2.05]	0.14 [1.83]
	(5)	0.42 [2.25]	0.50 [2.91]	0.56 [3.75]	0.66 [4.26]	0.63 [3.63]	0.22 [2.45]	0.26 [3.00]	0.25 [2.85]	0.24 [2.70]	0.29 [3.30]	0.29 [3.15]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CEI Quintiles					CEI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	258	259	260	256	254	24	24	22	21	22	
	(2)	79	79	79	79	78	42	42	43	42	43	
	(3)	61	61	61	60	60	78	78	77	78	78	
	(4)	55	55	55	55	55	182	177	181	178	177	
(5)	52	52	52	52	51	1116	1324	1860	1507	1258		

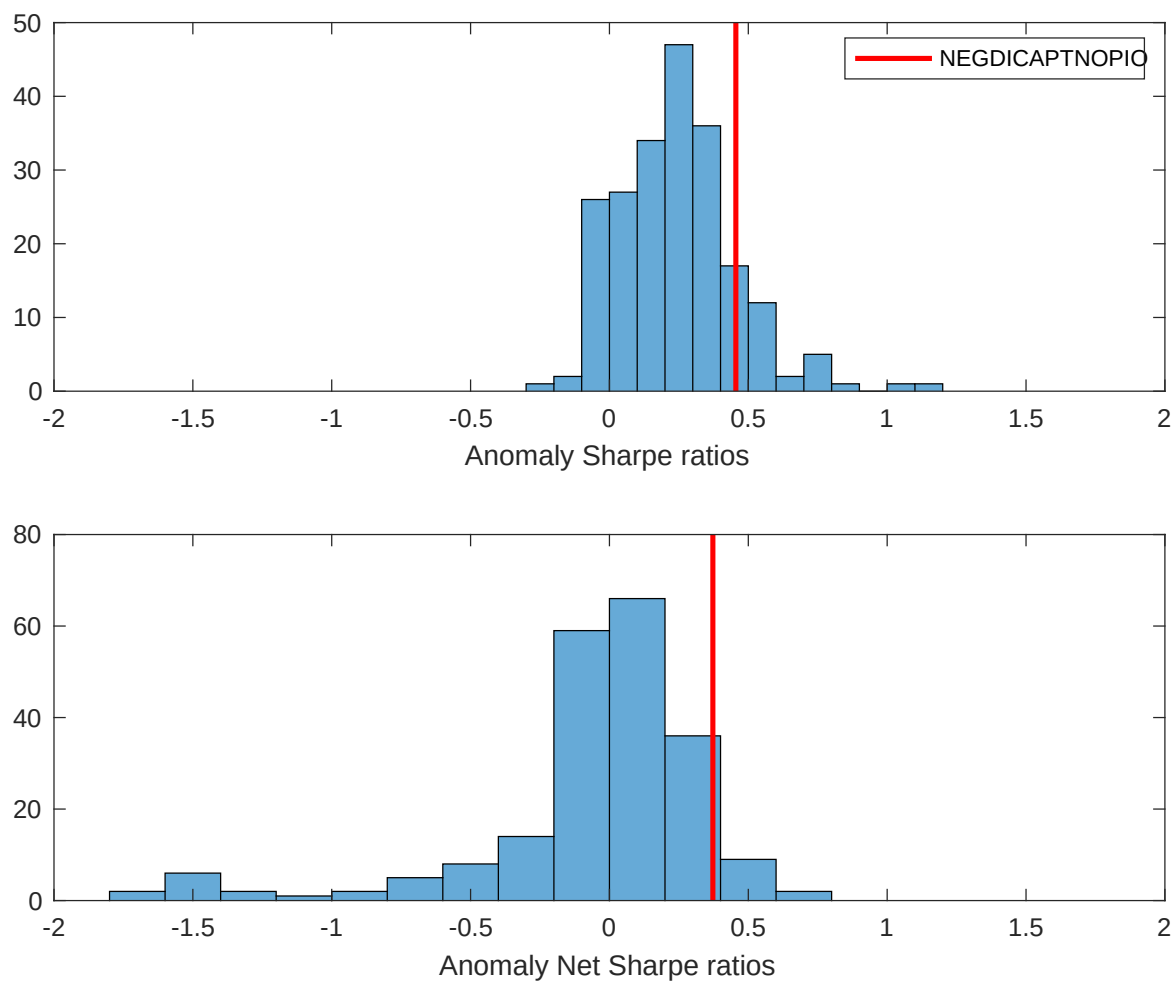


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

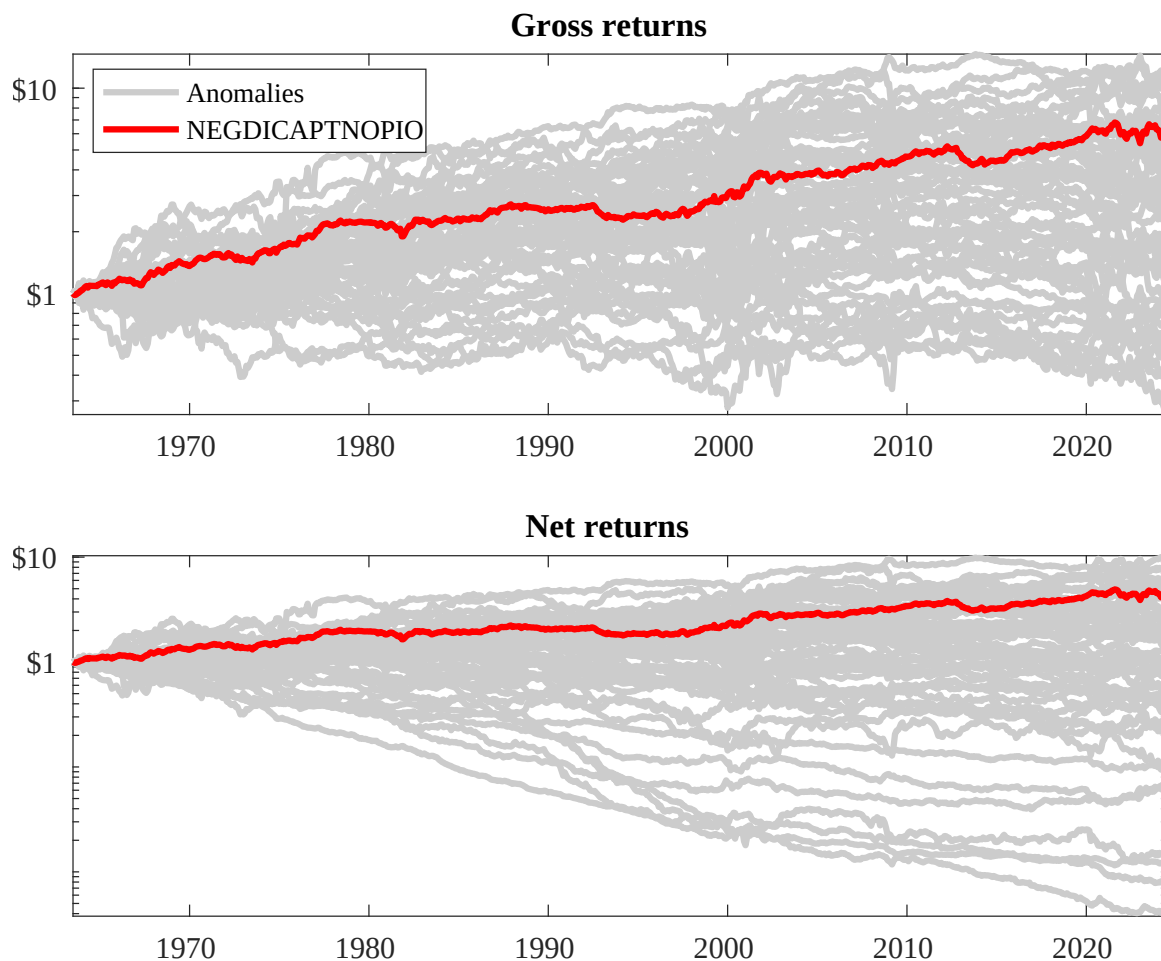
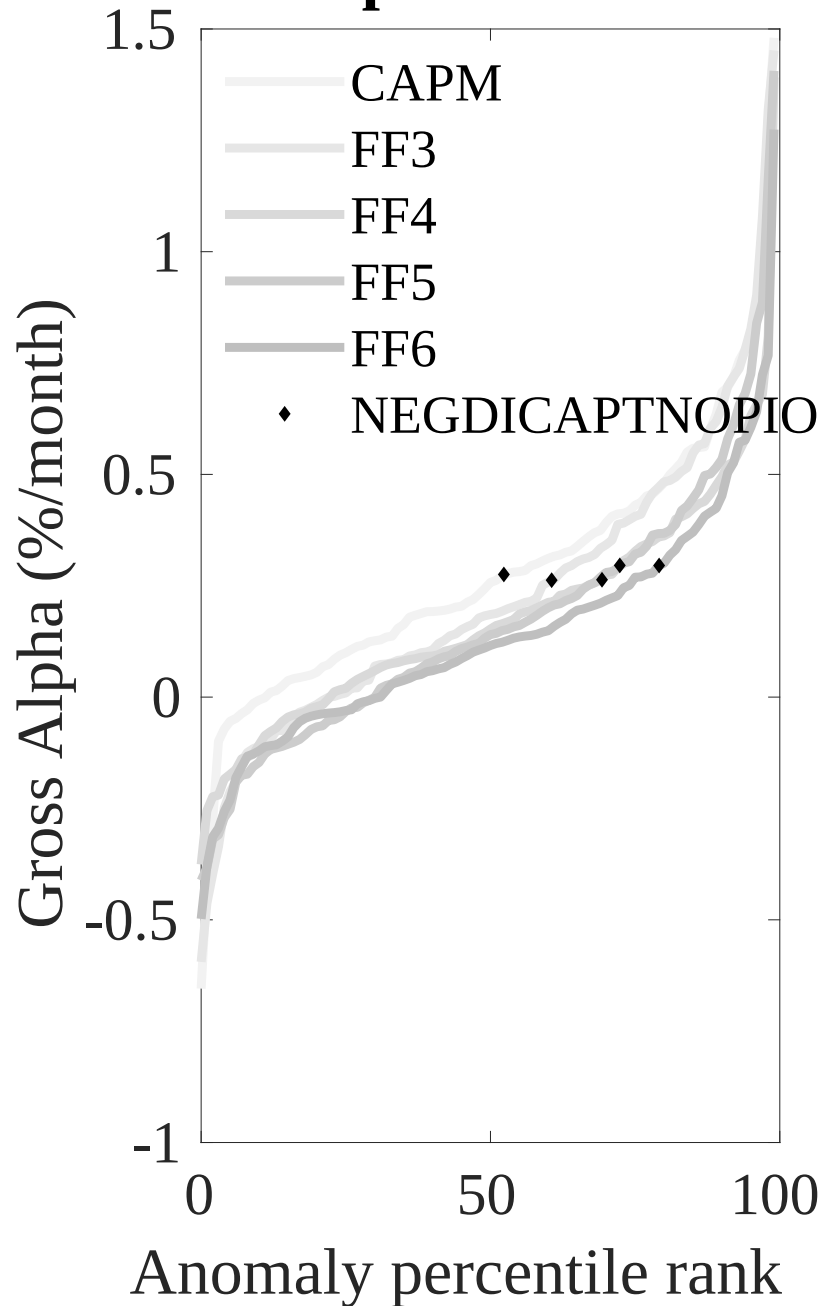


Figure 3: Dollar invested.

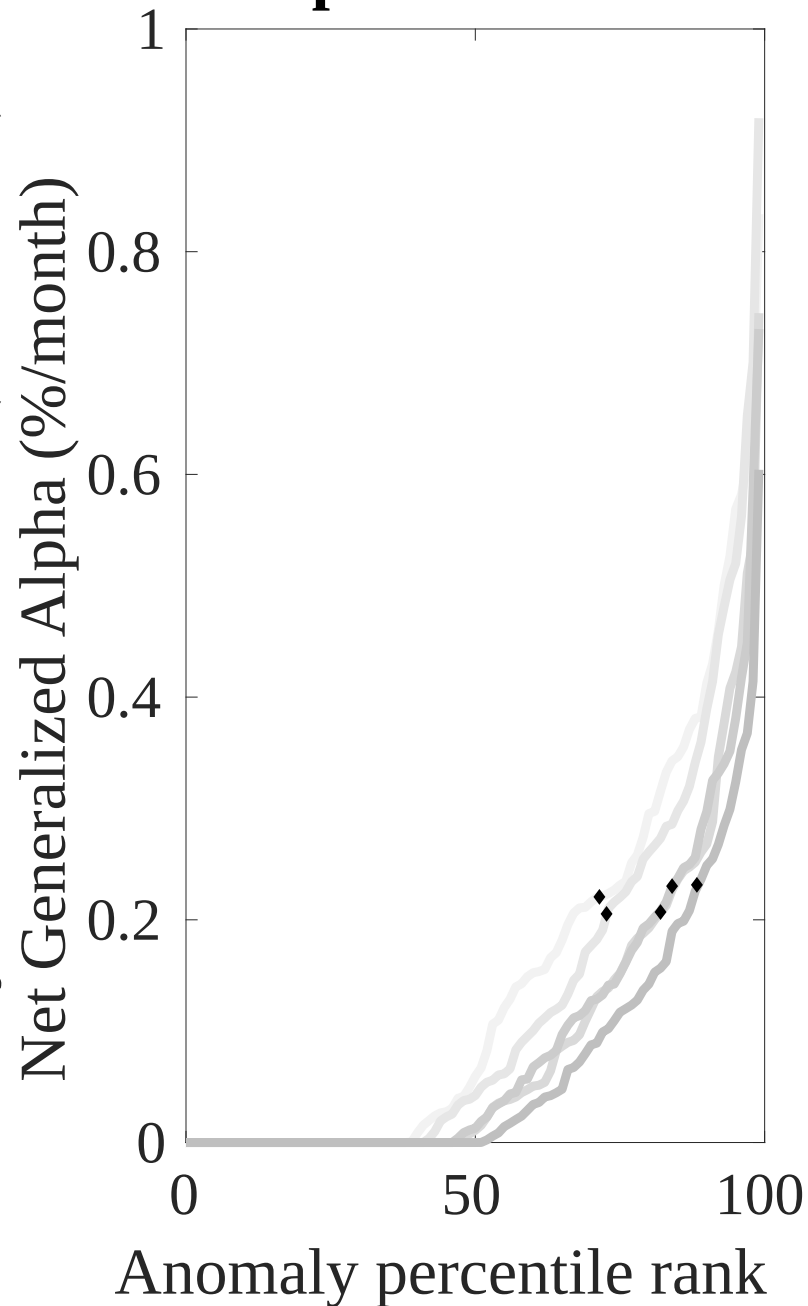
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



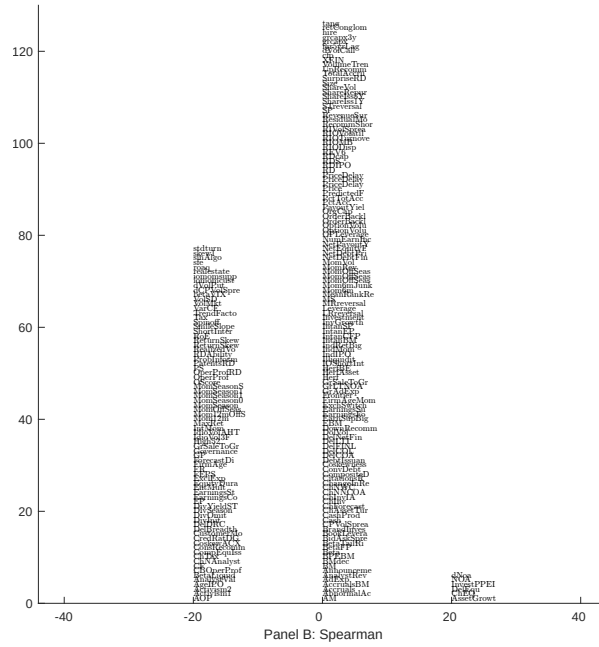
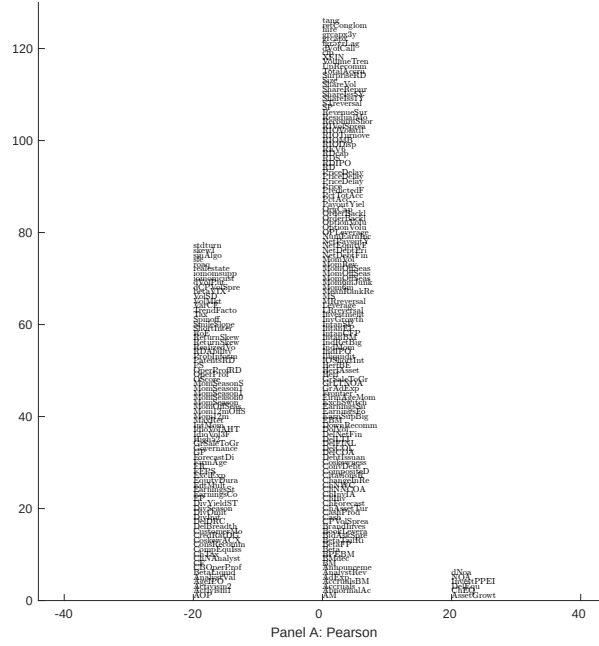


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CEI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

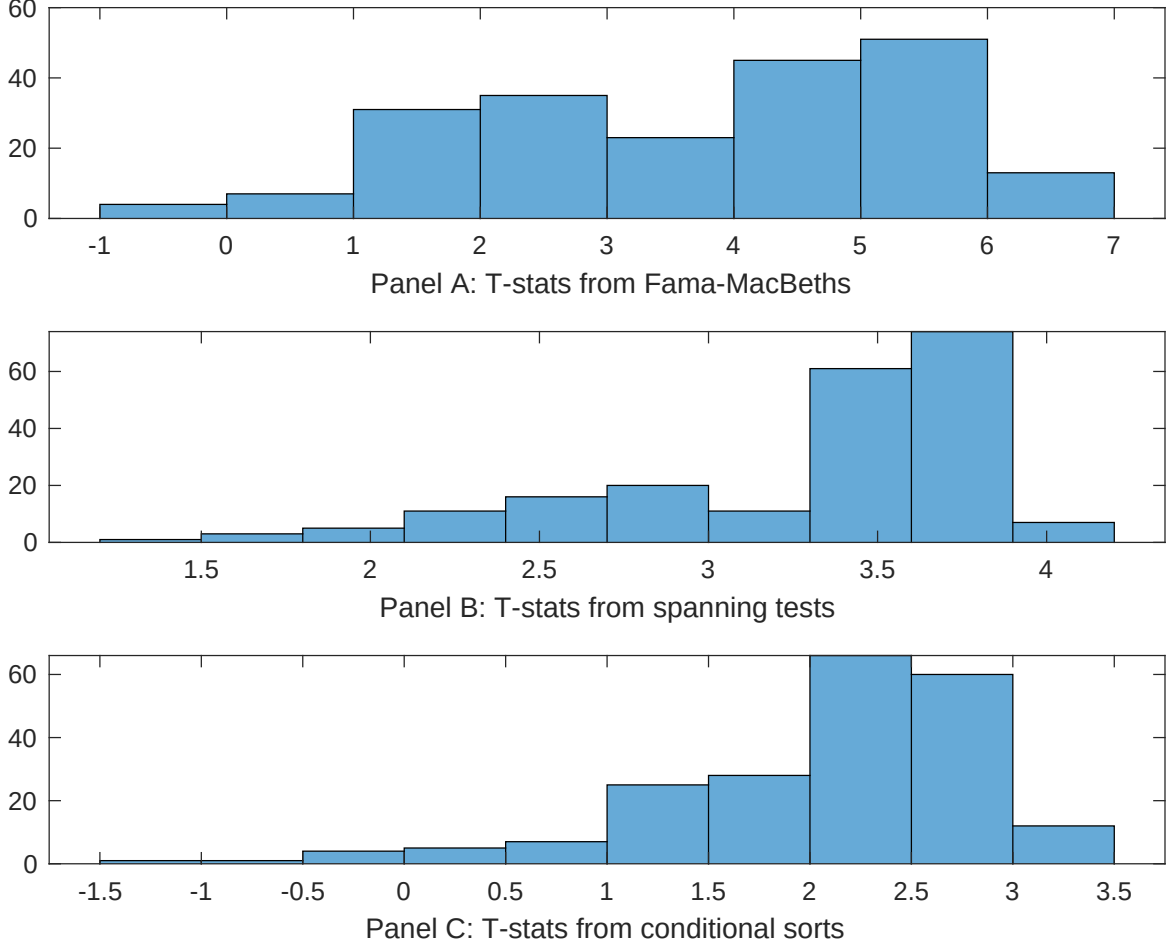


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CEI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [6.82]	0.13 [6.03]	0.12 [5.58]	0.13 [5.67]	0.13 [5.82]	0.17 [7.02]	0.15 [5.03]
CEI	0.37 [4.24]	0.24 [2.89]	0.41 [4.56]	0.34 [3.22]	0.16 [1.86]	0.29 [3.46]	0.95 [0.93]
Anomaly 1	0.41 [3.57]						0.81 [0.72]
Anomaly 2		0.16 [7.83]					0.43 [1.43]
Anomaly 3			0.29 [5.58]				0.28 [0.41]
Anomaly 4				0.49 [8.13]			0.94 [1.30]
Anomaly 5					0.13 [9.13]		0.61 [2.43]
Anomaly 6						0.83 [6.85]	0.19 [1.08]
# months	720	732	732	732	720	732	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CEI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CEI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28 [3.84]	0.29 [3.93]	0.26 [3.59]	0.29 [3.91]	0.28 [3.80]	0.26 [3.57]	0.24 [3.34]
Anomaly 1	12.12 [3.06]						10.23 [2.53]
Anomaly 2		7.91 [2.31]					2.73 [0.70]
Anomaly 3			12.85 [3.57]				12.55 [2.76]
Anomaly 4				6.09 [2.06]			-2.53 [-0.67]
Anomaly 5					3.80 [0.92]		-5.74 [-1.26]
Anomaly 6						9.68 [3.14]	8.90 [2.70]
mkt	-3.43 [-1.96]	-3.93 [-2.25]	-3.42 [-1.96]	-3.91 [-2.24]	-3.81 [-2.17]	-4.18 [-2.39]	-3.45 [-1.98]
smb	-1.15 [-0.45]	-1.06 [-0.42]	0.81 [0.32]	-0.34 [-0.13]	-0.70 [-0.28]	0.80 [0.31]	1.51 [0.58]
hml	-0.20 [-0.06]	0.43 [0.13]	0.44 [0.13]	1.00 [0.29]	1.00 [0.30]	-1.32 [-0.38]	-2.94 [-0.85]
rmw	-12.84 [-3.77]	-13.46 [-3.95]	-11.22 [-3.26]	-12.70 [-3.70]	-13.39 [-3.91]	-14.43 [-4.23]	-12.08 [-3.50]
cma	-5.11 [-0.82]	0.56 [0.10]	-1.78 [-0.32]	1.64 [0.30]	3.80 [0.65]	6.48 [1.31]	-7.38 [-1.06]
umd	-0.19 [-0.11]	0.01 [0.01]	-0.75 [-0.43]	-0.63 [-0.35]	-0.05 [-0.03]	-0.32 [-0.19]	-0.86 [-0.49]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	4	4	5	4	3	4	6

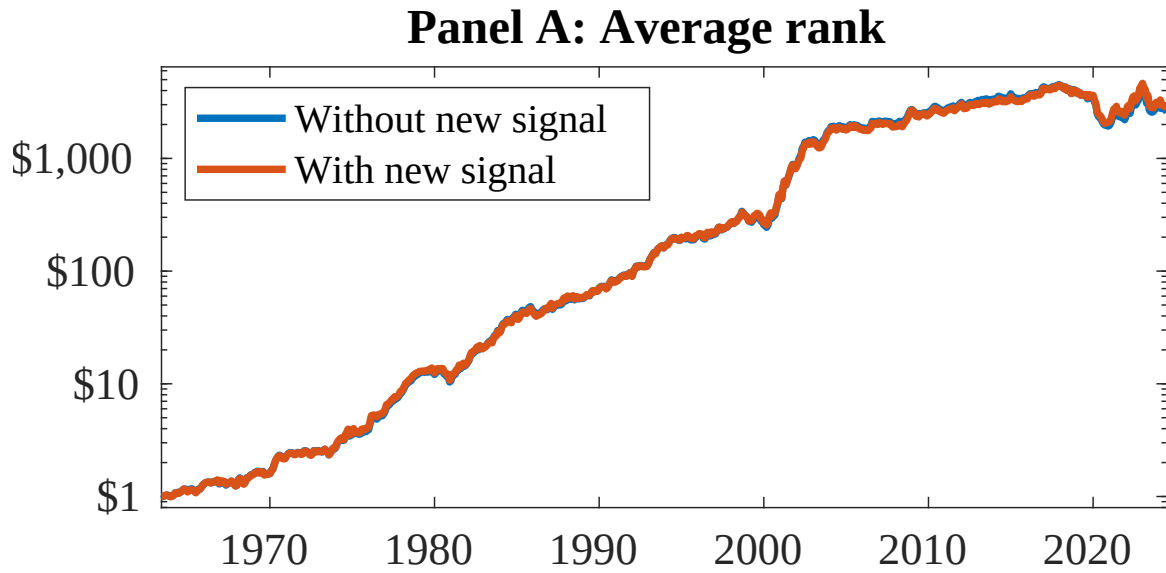


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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